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Introduction to Entropy

Seminar Stochastic Processes

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Abstract: This seminar paper provides a foundational overview of information theory, focusing on entropy, mutual information, and relative entropy. We establish key definitions and prove fundamental inequalities using Jensen's inequality and the log-sum inequality. To build intuition, theoretical results are complemented by numerical simulations of discrete random variables. Finally, we derive the relationship between relative entropy and cross-entropy, demonstrating their practical application in optimizing neural networks for the MNIST digit classification task.

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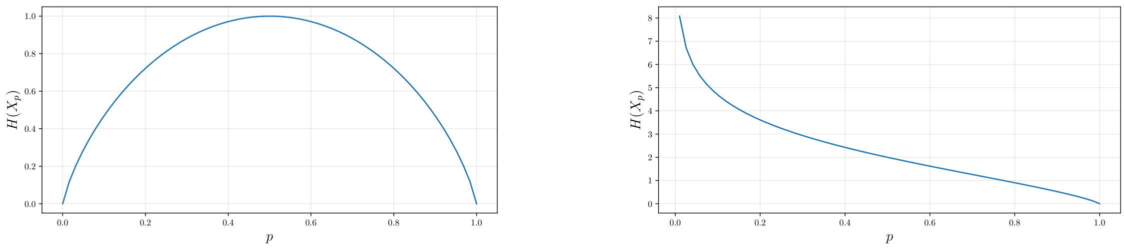


(a) Bernoulli Rand. Variable Entropy $H(X_p)$

(b) Geometric Rand. Variable Entropy $H(X_p)$

1 Entropy and Mutual information

1.1 Definitions and Conventions

Definition 1. Let $(\Omega, \mathcal{A}, \mathbb{P})$ be a probability space, let $\mathcal{X}$ be a countable set and let $X : \Omega \to \mathcal{X}$ be a discrete random variable on $(\Omega, \mathcal{A}, \mathbb{P})$. We can then define

$$\textbf{Entropy wrt. base:} \quad H_b(X) = \mathbb{E}\left(-\log_b p_X(X)\right) = -\sum_{x \in \operatorname{supp}(X)} p_X(x) \log_b p_X(x)$$

$$\textbf{Entropy conventionally:} \quad H(X) = H_2(X)$$

Remark 1.

1. Let $\mathcal{X}, \mathcal{Y}$ be countable sets and $X : \Omega \to \mathcal{X}$, $Y : \Omega \to \mathcal{Y}$ be discrete random variables on $(\Omega, \mathcal{A}, \mathbb{P})$. From now on, the random variables X, Y are always available for use.

2. The shorthand notations $p(x) = \mathbb{P}[X = x]$ and $p(y) = \mathbb{P}[Y = y]$ from [CT05] exist. We do **not** use this bare shorthand in this paper. Instead, we use probability mass function notation $p_X(x) = \mathbb{P}[X = x]$, $p_Y(y) = \mathbb{P}[Y = y]$, and $p_{X,Y}(x, y) = \mathbb{P}[X = x, Y = y]$. For conditional probabilities, we write $p_{Y|X}(y \mid x) = \mathbb{P}[Y = y \mid X = x]$ (for $\mathbb{P}[X = x] > 0$).

3. For expectation with respect to a specific probability measure (equivalently, in a discrete setting, its probability mass function), we use them as subscripts on $\mathbb{E}$, for example $\mathbb{E}_{p_X}$, $\mathbb{E}_{p_{X,Y}}$, or more generally $\mathbb{E}_p$. If no subscript is given, expectation is taken with respect to the ambient measure $\mathbb{P}$.

4. We use the convention $\log = \log_2$, as the entropy H is defined wrt. base 2.

5. We also use the following convention and justify it through a continuity argument:

$$0 \log 0 = \lim_{x \to 0^+} x \log x = \lim_{x \to 0^+} \frac{\log x}{\frac{1}{x}} = \lim_{x \to 0^+} \frac{\frac{1}{\ln(2)x}}{-\frac{1}{x^2}} = \lim_{x \to 0^+} \frac{-x}{\ln(2)} = 0$$

   This choice is sensible, as $\log x$ is not defined for negative x.

6. The conventions, definitions and theorems are from Definitions, Theorems, Remarks and Exercises in *Elements of Information Theory, second edition* (see [CT05]).

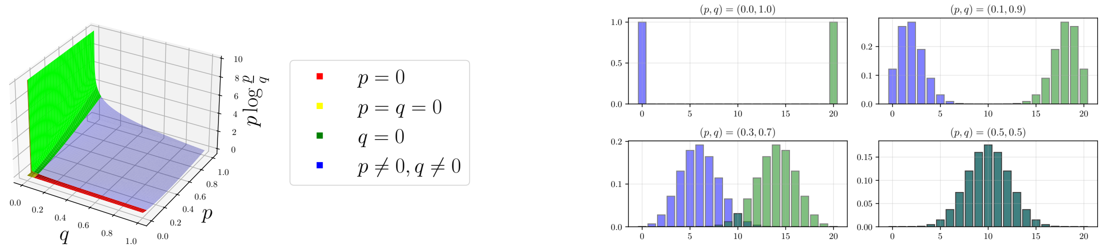
Remark 2 (Existence of Entropy)**.** Note that if $|\mathcal{X}|$ is finite, $(\forall b \in \mathbb{R}_+ : H_b(X)$ finite) and $H(X) \leq \log |\mathcal{X}|$ (see Theorem 5). For $|\mathcal{X}|$ countably infinite, there are counterexamples where $H_b(X) = \infty$ (see [Mat]). From now on, we will assume that entropy is finite.

Example 1 (Entropy of bernoulli variable)**.** Let $p \in (0, 1)$ and $X_p \sim B(1, p)$ be a weighted coin flip. We can calculate the Entropy of X_p: $H(X_p) = -p \log p - (1-p) \log(1-p)$. A visual inspection (see Figure 1a) reveals that $H(X_p)$ seems to be maximised for $p = 0.5$ and minimised for $p \in \{0, 1\}$. An increase in uncertainty about the result of the coin flip seems to correspond with an increase in entropy.

Example 2 (Entropy of geometric variable)**.** Let $p \in (0, 1)$ and $X_p \sim G(p)$ be the number of times a weighted coin is flipped, until the first head occurs. We will calculate the Entropy of X_p. It will require the two well-known series:

$$\forall r \in (0, 1) : \sum_{n \in \mathbb{N}_0} r^n = \frac{1}{1 - r} \tag{1}$$

$$\forall r \in (0, 1) : \sum_{n \in \mathbb{N}_0} n r^n = \frac{r}{(1 - r)^2} \tag{2}$$



(a) Pointwise Relative Entropy

(b) Relative Entropies of Binomial Distributions

We can now directly calculate the Entropy of X_p:

$$
\begin{aligned}
H(X_p) &= \sum_{x\in\mathbb{N}} -p_{X_p}(x)\log p_{X_p}(x) && \text{(Def. of Entropy)}\\
&= -\sum_{x\in\mathbb{N}} (1-p)^{x-1}p\log\left((1-p)^{x-1}p\right) && \text{(Subst. in geometric mass function)}\\
&= -\sum_{x\in\mathbb{N}} (1-p)^{x-1}p\left((x-1)\log(1-p)+\log p\right) && \text{(Log rules)}\\
&= -p\log(1-p)\sum_{x\in\mathbb{N}_0}\left((1-p)^x x\right) - p\log p\sum_{x\in\mathbb{N}_0}\left((1-p)^x\right) && \text{(Factor out constants)}\\
&= -p\log(1-p)\frac{1-p}{(1-(1-p))^2} - p\log p\frac{1}{1-(1-p)} && \text{(Use series 1 and 2)}\\
&= -p\log(1-p)\frac{1-p}{p^2} - p\log p\frac{1}{p} && \text{(Simplify expr.)}\\
&= \frac{-(1-p)\log(1-p)-p\log p}{p} && \text{(Simplify expr.)}
\end{aligned}
$$

For $p = 0.5$ we get $H(X_{0.5}) = \frac{-0.5\log 0.5 - 0.5\log 0.5}{0.5} = -2\log 0.5 = 2$.
We can visually inspect $(0,1)\to\mathbb{R}, p\mapsto H(X_p)$ (see Figure 1b) to get a feeling for the entropy of X_p. An increase in p is linked to lower variance and more concentration of the distribution towards zero. Based on the plot, that increase looks to be linked to a lower entropy and vice-versa.

We will now find a strategy, that calculates the number of flips until the first head occurs using simple yes/no questions. A simple strategy is to ask "Is $X = 1$?", "Is $X = 2$?" and so on. Under this strategy, the number of questions required is exactly the value of X. Thus, the average number of questions is simply $\mathbb{E}(X_{0.5}) = \frac{1}{0.5} = 2$. This matches the entropy $H(X_{0.5}) = 2$.

Definition 2. We define the following

$$
\begin{aligned}
\textbf{Conditional Entropy:}\quad & H(X \mid Y) = -\mathbb{E}\left(\log p_{X|Y}(X \mid Y)\right)\\
\textbf{Joint Entropy:}\quad & H(X,Y) = -\mathbb{E}\left(\log p_{X,Y}(X,Y)\right)
\end{aligned}
$$

Let p and q be two probability mass functions on the same set $\mathcal{Z}$ and let $X \sim p$. We define the following:

$$\textbf{Relative Entropy:}\quad D(p\|q) = \mathbb{E}_p\left(\log\frac{p(X)}{q(X)}\right) \text{ with conventions from Remark 3}$$

(Sometimes also called KL-Divergence)

Using Relative Entropy, we can define the Mutual Information between the variables X, Y:

$$\textbf{Mutual Information:}\quad I(X;Y) = D(p_{X,Y}\|p_X\,p_Y)$$

Remark 3. We have

$$
\begin{aligned}
D(p\|q) &= \mathbb{E}_p\left(\log\frac{p(X)}{q(X)}\right) && \text{(def. of relative entropy)}\\
&= \sum_{x\in\mathcal{X}} p(x)\log\frac{p(x)}{q(x)} && \text{(def. of expected value)}
\end{aligned}
$$

To understand the conventions, we can look at the limit cases:

1. Case $p \in (0,1], q = 0$: $\lim_{q\to 0^+} p \log \frac{p}{q} = \lim_{q\to 0^+} (p\log p - p \log q) = \infty$.

2. Case $p = 0, q \in (0,1]$: $0 \log \frac{0}{q} = 0$.

3. Case $p = q = 0$: Case 1 logic yields $\lim_{q\to 0^+} p \log \frac{p}{q} = \infty$ and Case 2 logic yields $0 \log \frac{0}{0} = 0$. So what do we choose? As we want $\sum_{x\in\mathcal{X}} p(x) \log \frac{p(x)}{q(x)}$ to sum over $x \in \mathcal{X}, p(x) > 0$, we choose the convention $0 \log \frac{0}{0} = 0$.

Figure 2a visualizes the pointwise relative entropy function $(p,q) \mapsto p \log \frac{p}{q}$.

Example 3. To understand the concept, we can now calculate the relative entropies for an example. Let $X \sim B(20,\alpha)$ and $Y \sim B(20,\beta)$ with $(\alpha,\beta) \in [0,1]^2$.

$$
\begin{aligned}
& D(p_X \| p_Y) = \sum_{x=0}^{20} p_X(x) \log \frac{p_X(x)}{p_Y(x)} \\
\alpha = 0, \beta = 1: \quad & D(p_X \| p_Y) = 1 \log \frac{1}{0} + 0 \log \frac{0}{1} = \infty + 0 = \infty \\
\alpha = 0.1, \beta = 0.9: \quad & D(p_X \| p_Y) \approx 50.7 \\
\alpha = 0.3, \beta = 0.7: \quad & D(p_X \| p_Y) \approx 9.8 \\
\alpha = 0.5, \beta = 0.5: \quad & D(p_X \| p_Y) = \sum_{x=0}^{20} p_X(x) \log 1 = 0
\end{aligned}
$$

Figure 2b visualizes the two discrete distribution functions in the cases above. Intuitively, the more overlap the distributions have, the closer to zero the relative entropy is.

1.2 Mutual Information and Chain Rules

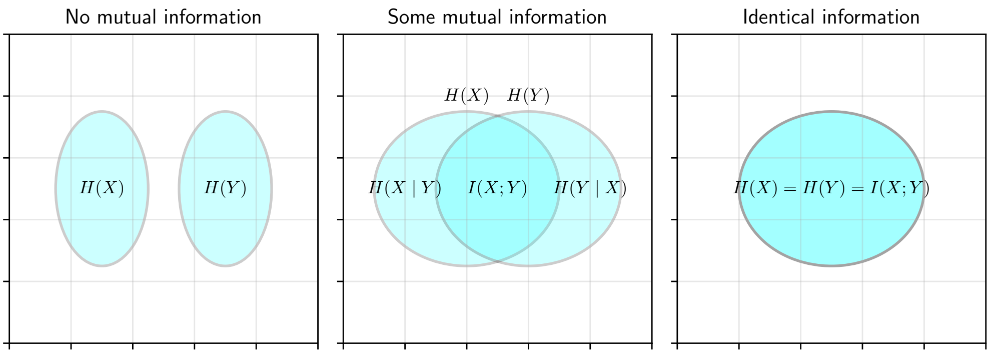


Figure 3: Relationship between Entropy, Conditional Entropy and Mutual Information

Theorem 1 (Chain Rule for Entropy)**.** *We have*

$$H(X,Y) = H(X) + H(Y \mid X)$$

Proof.

$$
\begin{aligned}
H(X,Y) &= -\mathbb{E}\left(\log p_{X,Y}(X,Y)\right) = -\mathbb{E}\left(\log p_{Y|X}(Y \mid X) p_X(X)\right) \\
&= -\mathbb{E}\left(\log p_{Y|X}(Y \mid X)\right) - \mathbb{E}\left(\log p_X(X)\right) = H(X) + H(Y \mid X)
\end{aligned}
$$

□

Theorem 2. *There are multiple equivalent ways to express Mutual Information (see Figure 3):*

1. $I(X;Y) = H(Y) - H(Y \mid X)$

2. $I(X;Y) = I(Y;X)$

3. $I(Y;X) = H(X) - H(X \mid Y)$

4. $I(X;Y) = H(X) + H(Y) - H(X,Y)$

5. $I(X;X) = H(X)$

Proof. 1. We can use the definition of mutual information and relative entropy to obtain:

$$\begin{aligned}
I(X;Y) &= D(p_{X,Y} \| p_X p_Y) && \text{(by def. of mutual info.)} \\
&= \mathbb{E}_{p_{X,Y}} \left(\log \frac{p_{X,Y}(X,Y)}{p_X(X) p_Y(Y)} \right) && \text{(by def. relative entropy)} \\
&= \mathbb{E}_{p_{X,Y}} \left(\log \frac{p_X(X) p_{Y|X}(Y \mid X)}{p_X(X) p_Y(Y)} \right) && \text{(using cond. probability)} \\
&= \mathbb{E}_{p_{X,Y}} \left(\log \frac{p_{Y|X}(Y \mid X)}{p_Y(Y)} \right) && \text{(simplify fraction)} \\
&= \mathbb{E}_{p_{X,Y}} \left(\log p_{Y|X}(Y \mid X) \right) - \mathbb{E}_{p_{X,Y}} \left(\log p_Y(Y) \right) && \text{(simplify logarithm)} \\
&= -H(Y \mid X) + H(Y) && \text{(by def. of entropy)}
\end{aligned}$$

2. The definition of mutual information yields:

$$\begin{aligned}
I(X;Y) &= D(p_{X,Y} \| p_X p_Y) && \text{(by def. of mutual info.)} \\
&= D(p_{Y,X} \| p_Y p_X) = I(Y;X)
\end{aligned}$$

3. Follows directly from 1 and 2.

4.

$$\begin{aligned}
I(X;Y) &= H(Y) - H(Y \mid X) && \text{(by 1)} \\
&= H(Y) - (H(X,Y) - H(X)) && \text{(chain rule)} \\
&= H(X) + H(Y) - H(X,Y)
\end{aligned}$$

5. Using the Defintion we get $H(X \mid X) = 0$. Using 1 we get $I(X;X) = H(X) - H(X \mid X) = H(X)$.

□

Definition 3. Let p and q be probability mass functions on $\mathcal{X} \times \mathcal{Y}$. We define the following:

Conditional Relative Entropy:

$$\begin{aligned}
& D(p_{Y|X} \| q_{Y|X}) \\
&= \mathbb{E}_{p_X} D(p_{Y|X}(. \mid X) \| q_{Y|X}(. \mid X)) \\
&= \sum_{x \in \mathcal{X}} p_X(x) D(p_{Y|X}(. \mid x) \| q_{Y|X}(. \mid x)) \\
&= \mathbb{E}_{p_{X,Y}} \left(\log \frac{p_{Y|X}(Y \mid X)}{q_{Y|X}(Y \mid X)} \right) \\
&= \sum_{(x,y) \in \mathcal{X} \times \mathcal{Y}} p_{X,Y}(x,y) \log \frac{p_{Y|X}(y \mid x)}{q_{Y|X}(y \mid x)}
\end{aligned}$$

Conditional Mutual Information: $I(X;Y \mid Z) = H(X \mid Z) - H(X \mid Y, Z)$

Example 4. The Conditional Relative Entropy can be illustrated using a Christmas example: Let $\Omega = \{1, \cdots, 12\}$, $\mathbb{P}$ uniformly distributed,
$\mathcal{X} = \{\textbf{JANUARY}, \cdots, \textbf{DECEMBER}\}$ and $\mathcal{Y} = \{0, 1\}$.
Our variable X gives us the month of the year:
We define it as $X : \Omega \to \mathcal{X}, \omega \mapsto$ **the corresponding month**.
Our variable Y answers the question "Is Christmas Advent?":
We define it as $Y : \Omega \to \mathcal{Y}, \omega \mapsto \mathbb{1}_{\{\textbf{NOVEMBER}, \textbf{DECEMBER}\}}(\omega)$.
These models are probability mass functions on $\mathcal{X} \times \mathcal{Y}$, with associated conditional probability mass functions $p_{Y|X}$ and $q_{Y|X}$. We have lived on earth our entire lives. Our model of reality p is correct! We define it as

$$p_{Y|X}(1 \mid x) = \begin{cases} 1, & \text{if } x \in \{\textbf{NOVEMBER}, \textbf{DECEMBER}\} \\ 0, & \text{otherwise} \end{cases}$$

The alien ET visits earth. He sees the Christmas trees in January and assumes Advent continues into January. In November and December, he is uncertain too. His model of reality q is not correct! We define it as

$$q_{Y|X}(1 \mid x) = \begin{cases} 0.5, & \text{if } x \in \{\mathbf{NOVEMBER}, \mathbf{DECEMBER}, \mathbf{JANUARY}\} \\ 0, & \text{otherwise} \end{cases}$$

Let us calculate the individual relative entropies:

1. $D(p_{Y|X}(\cdot \mid \mathbf{NOV.}) \| q_{Y|X}(\cdot \mid \mathbf{NOV.})) = 0 + 1 \log \frac{1}{0.5} = 1$
2. $D(p_{Y|X}(\cdot \mid \mathbf{DEC.}) \| q_{Y|X}(\cdot \mid \mathbf{DEC.})) = 0 + 1 \log \frac{1}{0.5} = 1$
3. $D(p_{Y|X}(\cdot \mid \mathbf{JAN.}) \| q_{Y|X}(\cdot \mid \mathbf{JAN.})) = 1 \log \frac{1}{0.5} + 0 = 1$
4. Otherwise: $D(p_{Y|X}(\cdot \mid x) \| q_{Y|X}(\cdot \mid x)) = 0$

We can now calculate the Conditional Relative Entropy:

$$\begin{aligned} D(p_{Y|X} \| q_{Y|X}) &= \sum_{x \in \mathcal{X}} p_X(x) D(p_{Y|X}(\cdot \mid x) \| q_{Y|X}(\cdot \mid x)) \\ &= \frac{1}{12}(1 + 1 + 1) = \frac{3}{12} = 0.25 \end{aligned}$$

So the expression calculates how much of an error ET makes per month, on average.

Theorem 3 (Chain Rules)**.** *Let $n \in \mathbb{N}, n \geq 2$, $(X_1, \cdots, X_n) \sim p(x_1, \cdots, x_n)$ and Y a random variable. The following statements about Entropy and Mutual Information are called Chain Rules:*

1.
$$H(X_1, \cdots, X_n) = \sum_{i=1}^{n} H(X_i \mid X_{i-1}, \cdots, X_1)$$

2.
$$D(p_{X,Y} \| q_{X,Y}) = D(p_X \| q_X) + D(p_{Y|X} \| q_{Y|X})$$

3.
$$I(X_1, \cdots, X_n; Y) = \sum_{i=1}^{n} I(X_i; Y \mid X_{i-1}, \cdots, X_1)$$

Proof. 1. Prove this result using induction by n.
Base case $n = 2$:

$$\begin{aligned} H(X_1, X_2) &= -\mathbb{E}(\log p(X_1, X_2)) \\ &= -\mathbb{E}(\log p(X_2 \mid X_1) p(X_1)) \\ &= -\mathbb{E}(\log p(X_2 \mid X_1)) + -\mathbb{E}(\log p(X_1)) \\ &= H(X_1) + H(X_2 \mid X_1) \end{aligned}$$

Assume the theorem holds for $n - 1$. Induction case $n - 1$ to n:

$$\begin{aligned} H(X_1, \cdots, X_n) &= H(X_n \mid X_1, \cdots, X_{n-1}) + H(X_1, \cdots, X_{n-1}) && \text{(apply base case)} \\ &= H(X_n \mid X_1, \cdots, X_{n-1}) + \sum_{i=1}^{n-1} H(X_i \mid X_{i-1}, \cdots, X_1) && \text{(induction hypothesis)} \end{aligned}$$

2. We prove the chain rule for relative entropy using a chain of equalities:

$$\begin{aligned} D(p_{X,Y} \| q_{X,Y}) &= \sum_{(x,y) \in \mathcal{X} \times \mathcal{Y}} p_{X,Y}(x, y) \log \frac{p_{X,Y}(x, y)}{q_{X,Y}(x, y)} \\ &= \sum_{(x,y) \in \mathcal{X} \times \mathcal{Y}} p_{X,Y}(x, y) \log \frac{p_{Y|X}(y \mid x) p_X(x)}{q_{Y|X}(y \mid x) q_X(x)} \\ &= \sum_{(x,y) \in \mathcal{X} \times \mathcal{Y}} p_{X,Y}(x, y) \log \frac{p_{Y|X}(y \mid x)}{q_{Y|X}(y \mid x)} + \sum_{(x,y) \in \mathcal{X} \times \mathcal{Y}} p_{X,Y}(x, y) \log \frac{p_X(x)}{q_X(x)} \\ &= D(p_{Y|X} \| q_{Y|X}) + D(p_X \| q_X) \end{aligned}$$

3. We prove the chain rule for mutual information using a chain of equalities:

$$\begin{aligned} I(X_1, \cdots, X_n; Y) &= H(X_1, \cdots, X_n) - H(X_1, \cdots, X_n \mid Y) \\ &= \sum_{i=1}^{n} H(X_i \mid X_{i-1}, \cdots, X_1) - \sum_{i=1}^{n} H(X_i \mid X_{i-1}, \cdots, X_1, Y) && \text{(using item 1)} \\ &= \sum_{i=1}^{n} I(X_i; Y \mid X_{i-1}, \cdots, X_1) && \text{(using item 1)} \end{aligned}$$

□

2 Inequalities for Entropy and Mutual Information

2.1 Convexity and Jensen's Inequality

Remark 4. We use the common definition of convex functions and concave functions from analysis.

Theorem 4 (Jensen's Inequality)**.** *Let $\mathcal{X} \subset \mathbb{R}$, $f : \mathcal{X} \to \mathbb{R}$ a function. Let $\mathbb{E}f(X)$ and $f(\mathbb{E}X)$ be defined.*

1. *If f is convex, we have $\mathbb{E}f(X) \geq f(\mathbb{E}X)$.*
2. *If f is concave, we have $\mathbb{E}f(X) \leq f(\mathbb{E}X)$.*
3. *If f is strictly convex and $\mathbb{E}f(X) = f(\mathbb{E}X)$, then $X = \mathbb{E}[X]$ almost surely.*

Proof.
1. Let $n \in \mathbb{N} \setminus \{1\}$ and, without loss of generality, assume that $p_1, \cdots, p_n \in (0,1)$ satisfy $\sum_{i=1}^{n} p_i = 1$ and $x_1, \cdots, x_n \in \mathbb{R}$. Distinguish two cases. If $\mathcal{X}$ is finite: We show $f\left(\sum_{i=1}^{n} p_i x_i\right) \leq \sum_{i=1}^{n} p_i f(x_i)$ by induction.
The definition of convexity yields the base case $i = 2$: $f(p_1 x_1 + p_2 x_2) \leq p_1 f(x_1) + p_2 f(x_2)$.
We assume the claim holds for $n-1$ and the induction case goes as follows:

$$\begin{aligned} f\left(\sum_{i=1}^{n} p_i x_i\right) &= f\left(p_1 x_1 + (1-p_1) \sum_{i=2}^{n} \frac{p_i}{1-p_1} x_i\right) \\ &\leq p_1 f(x_1) + (1-p_1) f\left(\sum_{i=2}^{n} \frac{p_i}{1-p_1} x_i\right) && \text{(def. of convexity)} \\ &\leq p_1 f(x_1) + (1-p_1) \sum_{i=2}^{n} \frac{p_i}{1-p_1} f(x_i) && \left(\text{induct. hypo. applies since } \sum_{i=2}^{n} \frac{p_i}{1-p_1} = 1\right) \\ &= \sum_{i=1}^{n} p_i f(x_i) \end{aligned}$$

Else: If $\mathcal{X}$ is countably infinite, the argument requires an additional approximation step. We omit this here and refer to *Elements of Information Theory* [CT05].

2. Follows from part 1 applied to $-f$.
3. If the support of X has more than one element, the inequality in part 1 is strict since f is strictly convex. This contradicts the assumption $\mathbb{E}f(X) = f(\mathbb{E}X)$. Hence $X = c$ almost surely for some constant c. By definition of the expectation, this constant must be $c = \mathbb{E}[X]$.

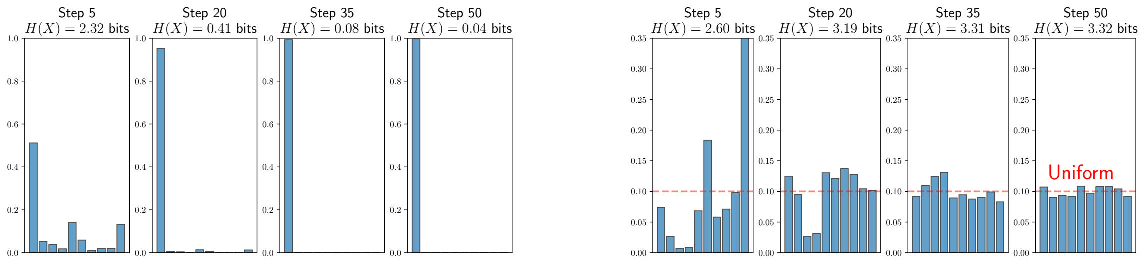
□

Corollary 1. *Entropy and Mutual Information are non-negative:*

1. $0 \leq H(X)$
2. $0 \leq D(p\|q)$
3. $0 \leq I(X;Y)$

Proof.
1. Note that $\log(\frac{1}{[0,1]}) = \log([1,\infty]) = [0,\infty]$ and $p_X(X) \in [0,1]$.
Using the monotonicity of the expected value, we obtain

$$0 \leq \mathbb{E}\left(\log\left(\frac{1}{p_X(X)}\right)\right) = -\mathbb{E}\left(\log\left(p_X(X)\right)\right) = H(X)$$



(a) Peak Distribution minimises Entropy

(b) Uniform Distribution maximises Entropy

2. We can prove this using Jensens Inequality on a *concave* function: Let $X \sim p$.

$$
\begin{aligned}
-D(p\|q) &= -\mathbb{E}_p\left(\log \frac{p(X)}{q(X)}\right) = \mathbb{E}_p\left(\log \frac{q(X)}{p(X)}\right) && \text{(def. of relative entropy)}\\
&\leq \log\left(\mathbb{E}_p \frac{q(X)}{p(X)}\right) && \text{(apply Jensen's inequality)}\\
&= \log\left(\sum_{x\in\mathrm{supp}(p)} p(x)\frac{q(x)}{p(x)}\right) = \log\left(\sum_{x\in\mathrm{supp}(p)} q(x)\right) && \text{(def. of exp. value, simplify expr.)}\\
&\leq \log\left(\sum_{x\in\mathcal{X}} q(x)\right) = \log(1) = 0 && \text{(q is a prob. function)}
\end{aligned}
$$

So $-D(p\|q) \leq 0$, hence $D(p\|q) \geq 0$.

3. Follows from part 2: $I(X;Y) = D(p_{X,Y}\|p_X p_Y) \geq 0$.

□

2.2 Additional Properties of Entropy

Remark 5. There are multiple natural questions we can ask about Entropy. We will look at an example for each of them and then prove the results in the follow-up theorem.

1. Can conditioning on more information increase the entropy?
Figure 3 illustrates the idea, that conditioning on more information can never increase the entropy. We will prove this exact statement: $H(X \mid Y) \leq H(X)$.

2. What distribution maximises the value of Entropy?
To get a computational intuition for this question, we need to limit our random variables to finite supports. Let $X_p \sim \mathrm{Categorical}(p_1, \cdots, p_n)$ be a categorical distribution.
We can now write the question as an optimisation problem:

$$\max_{p\in\mathbb{R}^n, \|p\|_1=1, p\geq 0} H(X_p)$$

The constraints can be automatically achieved by passing the logits through a *Softmax* function. We start from a random initialisation. Figure 4b illustrates the process of solving the problem using Gradient Descent using Tinygrad [Tin]. From the data, it seems plausible that the uniform distribution maximises entropy. We will prove this statement.

3. What distribution minimises the value of Entropy?
Figure 4a suggests that a peaked distribution minimises entropy. The setup is analogous to the previous optimisation problem and the figure was generated in the same way using Tinygrad. We will prove the statement

$$H(X) = 0 \iff \exists x \in \mathcal{X} : p_X(x) = 1.$$

4. What happens to the Entropy if we add independent noise to our measurements?
Let $X \sim U(\{1,2,3,4\})$ be the original signal, $N \sim U(\{0,1\})$ the noise and let X, N be independent. Then $S := X + N$ is a noisy signal. We expect the entropy to never decrease. We will prove this later.

Theorem 5. *We can formalise the previous observations including some more:*

1. *More information can only decrease entropy:*
$$H(X \mid Y) \leq H(X)$$

2. *A peaked distribution minimises entropy:*
$$H(X) = 0 \iff \exists x \in \mathcal{X} : p_X(x) = 1$$

3. *The uniform distribution maximizes entropy:*
$$H(X) \leq \log |\mathcal{X}| \text{ and } H(X) = \log |\mathcal{X}| \iff X \sim U(\mathcal{X})$$

4. *If Y is determined by X, then:*
$$H(Y \mid X) = 0 \implies \exists f : Y = f(X) \text{ almost surely}$$

5. *If independent noise is added to a random variable, entropy can only increase: Set $Z = X+Y$. Then we have*
$$X, Y \text{ independent} \implies H(X) \leq H(Z) \wedge H(Y) \leq H(Z)$$

6. *The joint entropy of many variables never exceeds the sum of the individual entropies:*
$$H(X_1, \cdots, X_n) \leq \sum_{i=1}^{n} H(X_i)$$

Proof. 1.
$$0 \leq I(X;Y) = H(X) - H(X \mid Y) \iff H(X \mid Y) \leq H(X)$$

2. We have
$$H(X) = \sum_{x \in \mathcal{X}} -p_X(x) \log p_X(x).$$
Each summand is non-negative, so $H(X) = 0$ holds if and only if
$$\forall x \in \mathcal{X} : -p_X(x) \log p_X(x) = 0.$$
For every $x \in \mathcal{X}$ with $p_X(x) > 0$, this implies $\log p_X(x) = 0$, hence $p_X(x) = 1$. Therefore, $H(X) = 0$ if and only if there exists $x \in \mathcal{X}$ with $p_X(x) = 1$.

3. Let $Y \sim U(\mathcal{X})$ be such that $\forall x \in \mathcal{X} : q(x) = \frac{1}{|\mathcal{X}|}$.
$$\begin{aligned}
0 \leq D(p_X \| q) &= \mathbb{E}_{p_X}\left(\log \frac{p_X(X)}{q(X)}\right) = \sum_{x \in \mathcal{X}} p_X(x) \log \frac{p_X(x)}{q(x)} \\
&= \sum_{x \in \mathcal{X}} p_X(x) \log\left(p_X(x)|\mathcal{X}|\right) = \sum_{x \in \mathcal{X}} p_X(x) \log p_X(x) + \log |\mathcal{X}| \sum_{x \in \mathcal{X}} p_X(x) \\
&= -H(X) + \log |\mathcal{X}| = \log |\mathcal{X}| - H(X)
\end{aligned}$$
This is equivalent to $H(X) \leq \log |\mathcal{X}|$. Lastly, the book Elements of Information Theory [CT05] outlines the use of Jensen's Inequality to prove that entropy is maximized only by the uniform distribution. Alternatively, it is more simple to use the theorem about Convexity of Entropy 7 (it comes later and is based on the Log-Sum Inequality), which confirms that $D(p_X \| q) = 0 \iff p_X = q$.

4. We have $H(Y \mid X) = -\mathbb{E}_{p_{X,Y}}\left(\log p_{Y|X}(Y \mid X)\right) = \sum_{(x,y) \in \mathcal{X} \times \mathcal{Y}} -p_{X,Y}(x,y) \log p_{Y|X}(y \mid x)$.
Additionally, we have $\forall (x,y) \in \mathcal{X} \times \mathcal{Y} : -p_{X,Y}(x,y) \log p_{Y|X}(y \mid x) \geq 0$.
Combining those facts, we get
$$\begin{aligned}
H(Y \mid X) = 0 &\iff \forall (x,y) \in \mathcal{X} \times \mathcal{Y} : p_{X,Y}(x,y) \log p_{Y|X}(y \mid x) = 0 \\
&\iff \forall (x,y) \in \mathcal{X} \times \mathcal{Y} : p_{X,Y}(x,y) = 0 \oplus p_{Y|X}(y \mid x) = 1
\end{aligned}$$
This tells us, that either $p_{X,Y}(x,y) = 0$ or $p_{X,Y}(x,y) = p_{Y|X}(y \mid x) p_X(x) = p_X(x) = 1$.
Define $(y_x)_{x \in \mathcal{X}}$ such that $\forall x \in \mathcal{X} : p_{X,Y}(x, y_x) > 0$.
Set $f : \operatorname{supp}(X) \to \mathcal{Y}, x \mapsto y_x$. This gets us $Im(f) = \{y_x : x \in \mathcal{X}\} = \{y \in \mathcal{Y} : p_{X,Y}(x,y) > 0\}$.
So $Y = f(X)$ almost surely.

5. We have

$$H(Z \mid X) = -\mathbb{E}_{p_{X,Y}}\left(\log p_{Z|X}(Z \mid X)\right) = \sum_{(x,y)\in\mathcal{X}\times\mathcal{Y}} -p_{X,Y}(x,y)\log\mathbb{P}[Z = x+y \mid X = x]$$
$$= \sum_{(x,y)\in\mathcal{X}\times\mathcal{Y}} -p_{X,Y}(x,y)\log\mathbb{P}[X+Y = x+y \mid X = x]$$
$$= \sum_{(x,y)\in\mathcal{X}\times\mathcal{Y}} -p_{X,Y}(x,y)\log\mathbb{P}[Y = y \mid X = x] = -\mathbb{E}_{p_{X,Y}}\left(\log p_{Y|X}(Y \mid X)\right) = H(Y \mid X)$$

Using Independence and Information never hurts, we get $H(Y) = H(Y \mid X) = H(Z \mid X) \leq H(Z)$. By an analogous argument, we get $H(X) = H(X \mid Y) = H(Z \mid Y) \leq H(Z)$.

6. Using the Chain rule, we have

$$\begin{aligned} H(X_1, \cdots, X_n) &= \sum_{i=1}^{n} H(X_i \mid X_1, \cdots, X_{i-1}) && \text{(chain rule)} \\ &\leq \sum_{i=1}^{n} H(X_i) && \text{(information never hurts)} \end{aligned}$$

□

2.3 The Log-Sum Inequality

Theorem 6 (Log-Sum Inequality)**.** *Let $n \in \mathbb{N}$, $a_1, \cdots, a_n, b_1, \cdots, b_n \geq 0$. Then we have*

$$\sum_{i=1}^{n} a_i \log\frac{a_i}{b_i} \geq \left(\sum_{i=1}^{n} a_i\right)\log\frac{\sum_{i=1}^{n} a_i}{\sum_{i=1}^{n} b_i}$$

with equality iff. $\exists c > 0, \forall i : a_i = cb_i$

Proof. The proof of this inequality goes beyond the scope of the paper. Please refer to Elements of Information Theory [CT05] for more information. □

Theorem 7 (Convexity of Entropy)**.** *1. $D(.\|.)$ is a convex function. This means that*

$$\begin{aligned} \forall p_1, q_1, p_2, q_2, \forall\lambda \in [0,1] : \quad & D(\lambda p_1 + (1-\lambda)p_2 \| \lambda q_1 + (1-\lambda)q_2) \\ & \leq \lambda D(p_1\|q_1) + (1-\lambda)D(p_2\|q_2) \end{aligned}$$

2. *Zero relative entropy implies equality:* $D(p\|q) = 0 \iff p = q$

3. *Let $X_p \sim B(1,p)$ for all $p \in (0,1)$ and $h(p) := H(X_p)$.*
Then $h \leq 1$ and h is a concave function.

Proof. 1. Let p_1, q_1, p_2, q_2 be probability mass functions on $\mathcal{X}$. Let $\lambda \in (0,1)$.
First of all, the set of probability densities is convex. So the above statement is well-defined. Secondly, the inequality needs to be verified:

$$\begin{aligned} & D(\lambda p_1 + (1-\lambda)p_2 \| \lambda q_1 + (1-\lambda)q_2) \\ &= \sum_{x\in\mathcal{X}} (\lambda p_1(x) + (1-\lambda)p_2(x))\log\frac{\lambda p_1(x) + (1-\lambda)p_2(x)}{\lambda q_1(x) + (1-\lambda)q_2(x)} && \text{(by definition)} \\ &\leq \sum_{x\in\mathcal{X}} \lambda p_1(x)\log\frac{p_1(x)}{q_1(x)} + (1-\lambda)p_2(x)\log\frac{p_2(x)}{q_2(x)} && \text{(log-sum inequality on the 2 two-sums)} \\ &= \lambda D(p_1\|q_1) + (1-\lambda)D(p_2\|q_2) && \text{(by definition)} \end{aligned}$$

2. Given $p = q$, since $\log 1 = 0$, we have $D(p\|q) = 0$. Given $D(p\|q) = 0$, we have

$$\begin{aligned} 0 = D(p\|q) &= \sum_{x\in\mathcal{X}} p(x)\log\frac{p(x)}{q(x)} \\ &\geq \left(\sum_{x\in\mathcal{X}} p(x)\right)\log\frac{\sum_{x\in\mathcal{X}} p(x)}{\sum_{x\in\mathcal{X}} q(x)} = \log\frac{\sum_{x\in\mathcal{X}} p(x)}{\sum_{x\in\mathcal{X}} q(x)} = \log 1 = 0 \end{aligned}$$

So the Log-Sum Inequality is sharp and $p \geq 0, q \geq 0$, which implies $p = cq$ with $c > 0$. Since both p and q are probability mass functions and sum to 1, $c = 1$ and $p = q$.

3. First, we have

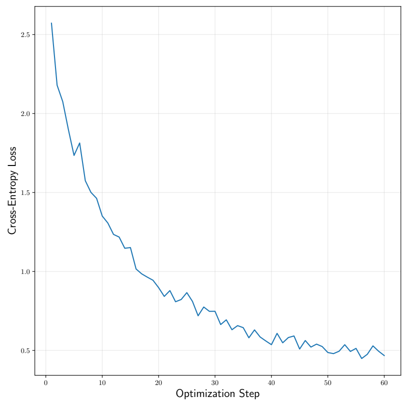
$$\begin{aligned}
h(p) = H(X_p) &= -p\log p - (1-p)\log(1-p) && \text{(by definition)}\\
&= -(p\log p + (1-p)\log(1-p)) && \text{(factor out)}\\
&\leq -(p+1-p)\log\frac{p+1-p}{1+1} = -\log\frac{1}{2} = 1 && \text{(simplify)}
\end{aligned}$$

Second, let $p_1, p_2 \in [0,1]$ and $\lambda \in (0,1)$. Using basic differentiation we get $h''(p) < 0$. That proves concavity:

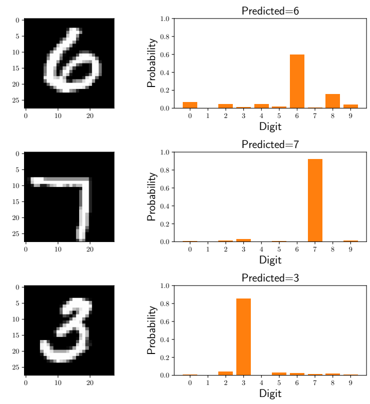
$$H(X_{\lambda p_1 + (1-\lambda)p_2}) \geq \lambda H(X_{p_1}) + (1-\lambda)H(X_{p_2})$$

□

2.4 Application to Optimisation



(a) MNIST objective function



(b) MNIST images and pred. digit distr.

Definition 4 (Cross-Entropy)**.** Let p, q be probability mass functions over the set $\mathcal{X}$ and $X \sim p$. Then Cross-Entropy is defined as

$$\mathrm{CrossEntropy}(p,q) = H(X) + D(p\|q)$$

Theorem 8. *1.*

$$\mathrm{argmin}_q\, \mathrm{CrossEntropy}(p,q) = \mathrm{argmin}_q\, D(p\|q)$$

2.

$$\mathrm{CrossEntropy}(p,q) = -\sum_{x\in\mathcal{X}} p(x)\log q(x)$$

Proof. 1. Cross-Entropy is simply Relative Entropy with a constant offset. So minimizing with respect to q yields the same minimizers.

2. Let $X \sim p$. Then we have

$$\begin{aligned}
\mathrm{CrossEntropy}(p,q) = H(X) + D(p\|q) &= \sum_{x\in\mathcal{X}} -p(x)\log p(x) + \sum_{x\in\mathcal{X}} p(x)\log\frac{p(x)}{q(x)}\\
&= \sum_{x\in\mathcal{X}} -p(x)\log p(x) + p(x)\log p(x) - p(x)\log q(x)\\
&= -\sum_{x\in\mathcal{X}} p(x)\log q(x)
\end{aligned}$$

□

Example 5 (MNIST Digit Classification)**.** We can now apply the concept of Relative Entropy to solve a common classification problem from machine learning. Let Ω be the set of 28x28 pixel images that contain exactly one handwritten digit from 0 to 9. The task is to predict the digit 0 to 9, based on the input image $\omega \in \Omega$.
In order to accomplish this task, we define a model family $(f_\phi)_{\phi\in\mathbb{R}^n}$ with $n \in \mathbb{N}$ and $f_\phi : \Omega \to \mathbb{R}^{10}_+$

for each $\phi \in \mathbb{R}^n$.
Each function f_ϕ outputs a probability mass function over digits. So $\forall \phi \in \mathbb{R}^n, \forall \omega \in \Omega : f_\phi(\omega) \geq 0 \wedge \sum_{i=1}^{10} f_\phi(\omega)_i = 1$.
Given training samples $\omega_1, \cdots, \omega_m \in \Omega$ with target distributions $d_{\omega_1}, \cdots, d_{\omega_m}$, the empirical optimisation objective is

$$\Phi = \text{argmin}_{\phi \in \mathbb{R}^n} \frac{1}{m} \sum_{i=1}^{m} D(d_{\omega_i} \| f_\phi(\omega_i))$$

which according to Theorem 8 is equivalent to

$$\Phi = \text{argmin}_{\phi \in \mathbb{R}^n} \frac{1}{m} \sum_{i=1}^{m} \text{CrossEntropy}(d_{\omega_i}, f_\phi(\omega_i))$$

As we have shown in Theorem 7, for fixed target distribution p, both Relative Entropy and Cross-Entropy are minimized at $q = p$.
For f_ϕ, we can use a two-layer convolutional neural network with dropout, as detailed in the official PyTorch Python example [Pyt]. The figure 5a is computed using the Tinygrad autograd library [Tin]. In this case, the map $\phi \mapsto f_\phi$ is not convex. So the optimisation objective is not convex either. But at least the per-sample loss function $\text{CrossEntropy}(d_{\omega_i}, .)$ is convex. We can now use Stochastic Gradient Descent to optimise for 60 steps with step size 5e-3 and batch size 512. Figure 5a illustrates the Value of the Objective Function over time and Figure 5b illustrates the outputs of the finished model.

References

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